$$C^{(3)}_{lq}\text{i1\_i2\_i3\_i4\_}\to$$

$$\begin{aligned}
&\frac{1}{16\pi^2}\Bigg(-\frac{1}{144}\,\frac{1}{m_\Phi^2}\Big(g_2^2\, c_\gamma^2\, \overline{y_u}^{i4p}\, y_u^{i3p}\,\delta_{i1i2} + s_\gamma^2\, \overline{y_d}^{i4p}\, y_d^{i3p}\left(-18\, c_\gamma^2\, \overline{y_e}^{i2r}\, y_e^{i1r} + g_2^2\,\delta_{i1i2}\right) + \\
&\qquad s_\gamma^2\, \overline{y_e}^{i2p}\, y_e^{i1p}\left(18\, c_\gamma^2\, \overline{y_u}^{i4r}\, y_u^{i3r} + g_2^2\,\delta_{i3i4}\right)\Big) + \\
&\frac{1}{9}\, g_2^4\, \mathrm{LF}_{3,0}[m_2]\,\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{6}\, g_2^4\, \mathrm{LF}_{4,-1}[m_2]\,\delta_{i1i2}\,\delta_{i3i4} - \frac{8}{45}\, g_2^4\, \mathrm{LF}_{5,-2}[m_2]\,\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{18}\sum_p g_2^4\, \mathrm{LF}_{3,0}\left[m_{\tilde l}^{p}\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{5}{48}\sum_p g_2^4\, \mathrm{LF}_{4,-1}\left[m_{\tilde l}^{p}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{2}{45}\sum_p g_2^4\, \mathrm{LF}_{5,-2}\left[m_{\tilde l}^{p}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{6}\sum_p g_2^4\, \mathrm{LF}_{3,0}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{5}{16}\sum_p g_2^4\, \mathrm{LF}_{4,-1}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{2}{15}\sum_p g_2^4\, \mathrm{LF}_{5,-2}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{4}\, s_\gamma^2\, c_\gamma^2\left(\overline{y_d}^{i4p}\, y_d^{i3p}\, \overline{y_e}^{i2r}\, y_e^{i1r} - \overline{y_e}^{i2p}\, y_e^{i1p}\, \overline{y_u}^{i4r}\, y_u^{i3r}\right)\mathrm{LF}_{1,2}[m_\Phi] + \\
&\frac{1}{24}\Big(-g_2^2\, c_\gamma^2\, \overline{y_u}^{i4p}\, y_u^{i3p}\,\delta_{i1i2} + s_\gamma^2\, \overline{y_d}^{i4p}\, y_d^{i3p}\left(3\, s_\gamma^2\, \overline{y_e}^{i2r}\, y_e^{i1r} - g_2^2\,\delta_{i1i2}\right) + \\
&\qquad s_\gamma^2\, \overline{y_e}^{i2p}\, y_e^{i1p}\left(3\, c_\gamma^2\, \overline{y_u}^{i4r}\, y_u^{i3r} - g_2^2\,\delta_{i3i4}\right)\Big)\,\mathrm{LF}_{2,1}[m_\Phi] + \\
&\frac{1}{72}\, g_2^2\left(3\left(s_\gamma^2\, \overline{y_d}^{i4p}\, y_d^{i3p} + c_\gamma^2\, \overline{y_u}^{i4p}\, y_u^{i3p}\right)\delta_{i1i2} + \left(3\, s_\gamma^2\, \overline{y_e}^{i2p}\, y_e^{i1p} + 4\, g_2^2\,\delta_{i1i2}\right)\delta_{i3i4}\right) \\
&\quad \mathrm{LF}_{3,0}[m_\Phi] - \frac{5}{48}\, g_2^4\, \mathrm{LF}_{4,-1}[m_\Phi]\,\delta_{i1i2}\,\delta_{i3i4} + \frac{2}{45}\, g_2^4\, \mathrm{LF}_{5,-2}[m_\Phi]\,\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{18}\, g_2^4\, \mathrm{LF}_{3,0}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{12}\, g_2^4\, \mathrm{LF}_{4,-1}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4} - \frac{4}{45}\, g_2^4\, \mathrm{LF}_{5,-2}[\tilde\mu]\,\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{96}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,1,0}\left[m_1, m_{\tilde l}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{96}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,2,-1}\left[m_1, m_{\tilde l}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{48}\, g_1^2\, g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_1, m_{\tilde l}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{96}\, g_1^2\, g_2^2\, \mathrm{LF}_{4,1,-2}\left[m_1, m_{\tilde l}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{96}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,1,0}\left[m_1, m_{\tilde l}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{96}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,2,-1}\left[m_1, m_{\tilde l}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{48}\, g_1^2\, g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_1, m_{\tilde l}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{96}\, g_1^2\, g_2^2\, \mathrm{LF}_{4,1,-2}\left[m_1, m_{\tilde l}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{864}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,1,0}\left[m_1, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{864}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,2,-1}\left[m_1, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{432}\, g_1^2\, g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_1, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{864}\, g_1^2\, g_2^2\, \mathrm{LF}_{4,1,-2}\left[m_1, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{864}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,1,0}\left[m_1, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{864}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,2,-1}\left[m_1, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{432}\, g_1^2\, g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_1, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{864}\, g_1^2\, g_2^2\, \mathrm{LF}_{4,1,-2}\left[m_1, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{96}\, g_2^4\, \mathrm{LF}_{2,1,0}\left[m_2, m_{\tilde l}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{96}\, g_2^4\, \mathrm{LF}_{2,2,-1}\left[m_2, m_{\tilde l}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{5}{48}\, g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_2, m_{\tilde l}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{32}\, g_2^4\, \mathrm{LF}_{4,1,-2}\left[m_2, m_{\tilde l}^{i1}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{96}\, g_2^4\, \mathrm{LF}_{2,1,0}\left[m_2, m_{\tilde l}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{96}\, g_2^4\, \mathrm{LF}_{2,2,-1}\left[m_2, m_{\tilde l}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{5}{48}\, g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_2, m_{\tilde l}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{32}\, g_2^4\, \mathrm{LF}_{4,1,-2}\left[m_2, m_{\tilde l}^{i2}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{96}\, g_2^4\, \mathrm{LF}_{2,1,0}\left[m_2, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{96}\, g_2^4\, \mathrm{LF}_{2,2,-1}\left[m_2, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{5}{48}\, g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_2, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{32}\, g_2^4\, \mathrm{LF}_{4,1,-2}\left[m_2, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{96}\, g_2^4\, \mathrm{LF}_{2,1,0}\left[m_2, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{96}\, g_2^4\, \mathrm{LF}_{2,2,-1}\left[m_2, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{5}{48}\, g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_2, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{32}\, g_2^4\, \mathrm{LF}_{4,1,-2}\left[m_2, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{18}\, g_2^2\, g_3^2\, \mathrm{LF}_{2,1,0}\left[m_3, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{18}\, g_2^2\, g_3^2\, \mathrm{LF}_{2,2,-1}\left[m_3, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{9}\, g_2^2\, g_3^2\, \mathrm{LF}_{3,1,-1}\left[m_3, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{18}\, g_2^2\, g_3^2\, \mathrm{LF}_{4,1,-2}\left[m_3, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{18}\, g_2^2\, g_3^2\, \mathrm{LF}_{2,1,0}\left[m_3, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{18}\, g_2^2\, g_3^2\, \mathrm{LF}_{2,2,-1}\left[m_3, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{9}\, g_2^2\, g_3^2\, \mathrm{LF}_{3,1,-1}\left[m_3, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{18}\, g_2^2\, g_3^2\, \mathrm{LF}_{4,1,-2}\left[m_3, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{48}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,1,0}\left[m_{\tilde l}^{i1}, m_1\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{96}\, g_1^2\, g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde l}^{i1}, m_1\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{48}\, g_2^4\, \mathrm{LF}_{2,1,0}\left[m_{\tilde l}^{i1}, m_2\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{96}\, g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde l}^{i1}, m_2\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{48}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,1,0}\left[m_{\tilde l}^{i2}, m_1\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{96}\, g_1^2\, g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde l}^{i2}, m_1\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{48}\, g_2^4\, \mathrm{LF}_{2,1,0}\left[m_{\tilde l}^{i2}, m_2\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{96}\, g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde l}^{i2}, m_2\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{432}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i3}, m_1\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{864}\, g_1^2\, g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i3}, m_1\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{48}\, g_2^4\, \mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i3}, m_2\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{96}\, g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i3}, m_2\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{9}\, g_2^2\, g_3^2\, \mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i3}, m_3\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{18}\, g_2^2\, g_3^2\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i3}, m_3\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{432}\, g_1^2\, g_2^2\, \mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i4}, m_1\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{864}\, g_1^2\, g_2^2\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i4}, m_1\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{48}\, g_2^4\, \mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i4}, m_2\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{96}\, g_2^4\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i4}, m_2\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{9}\, g_2^2\, g_3^2\, \mathrm{LF}_{2,1,0}\left[m_{\tilde q}^{i4}, m_3\right]\delta_{i1i2}\,\delta_{i3i4} + \frac{1}{18}\, g_2^2\, g_3^2\, \mathrm{LF}_{3,1,-1}\left[m_{\tilde q}^{i4}, m_3\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{8}\, g_2^2\, \overline{y_d}^{i4p}\, y_d^{i3p}\, \mathrm{LF}_{3,1,-1}\left[\tilde\mu, m_{\tilde d}^{p}\right]\delta_{i1i2} + \frac{1}{24}\, g_2^2\, \overline{y_d}^{i4p}\, y_d^{i3p}\, \mathrm{LF}_{4,1,-2}\left[\tilde\mu, m_{\tilde d}^{p}\right]\delta_{i1i2} - \\
&\frac{1}{8}\, g_2^2\, \overline{y_e}^{i2p}\, y_e^{i1p}\, \mathrm{LF}_{3,1,-1}\left[\tilde\mu, m_{\tilde e}^{p}\right]\delta_{i3i4} + \frac{1}{24}\, g_2^2\, \overline{y_e}^{i2p}\, y_e^{i1p}\, \mathrm{LF}_{4,1,-2}\left[\tilde\mu, m_{\tilde e}^{p}\right]\delta_{i3i4} - \\
&\frac{1}{8}\, g_2^2\, \overline{y_u}^{i4p}\, y_u^{i3p}\, \mathrm{LF}_{3,1,-1}\left[\tilde\mu, m_{\tilde u}^{p}\right]\delta_{i1i2} + \frac{1}{24}\, g_2^2\, \overline{y_u}^{i4p}\, y_u^{i3p}\, \mathrm{LF}_{4,1,-2}\left[\tilde\mu, m_{\tilde u}^{p}\right]\delta_{i1i2} + \\
&\frac{1}{16}\, g_2^4\, \mathrm{LF}_{2,1,1,-1}\left[m_2, m_{\tilde l}^{i1}, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{8}\, g_2^4\, m_2^2\, \mathrm{LF}_{2,1,1,0}\left[m_2, m_{\tilde l}^{i1}, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{16}\, g_2^4\, \mathrm{LF}_{2,1,1,-1}\left[m_2, m_{\tilde l}^{i2}, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} - \frac{1}{8}\, g_2^4\, m_2^2\, \mathrm{LF}_{2,1,1,0}\left[m_2, m_{\tilde l}^{i2}, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} + \\
&\frac{1}{8}\, \overline{y_d}^{i4p}\, y_d^{i3p}\, \overline{y_e}^{i2r}\, y_e^{i1r}\, \mathrm{LF}_{2,1,1,-1}\left[\tilde\mu, m_{\tilde d}^{p}, m_{\tilde e}^{r}\right] - \frac{1}{4}\, \tilde\mu^2\, \overline{y_e}^{i2p}\, y_e^{i1p}\, \overline{y_u}^{i4r}\, y_u^{i3r} \\
&\quad \mathrm{LF}_{2,1,1,0}\left[\tilde\mu, m_{\tilde e}^{p}, m_{\tilde u}^{r}\right] - \frac{1}{48}\, g_1^2\, g_2^2\, \mathrm{LF}_{1,1,1,1,-1}\left[m_1, m_2, m_{\tilde l}^{i1}, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{24}\, m_1\, m_2\, g_1^2\, g_2^2\, \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_2, m_{\tilde l}^{i1}, m_{\tilde q}^{i3}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{48}\, g_1^2\, g_2^2\, \mathrm{LF}_{1,1,1,1,-1}\left[m_1, m_2, m_{\tilde l}^{i2}, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4} - \\
&\frac{1}{24}\, m_1\, m_2\, g_1^2\, g_2^2\, \mathrm{LF}_{1,1,1,1,0}\left[m_1, m_2, m_{\tilde l}^{i2}, m_{\tilde q}^{i4}\right]\delta_{i1i2}\,\delta_{i3i4}\Bigg)
\end{aligned}$$